

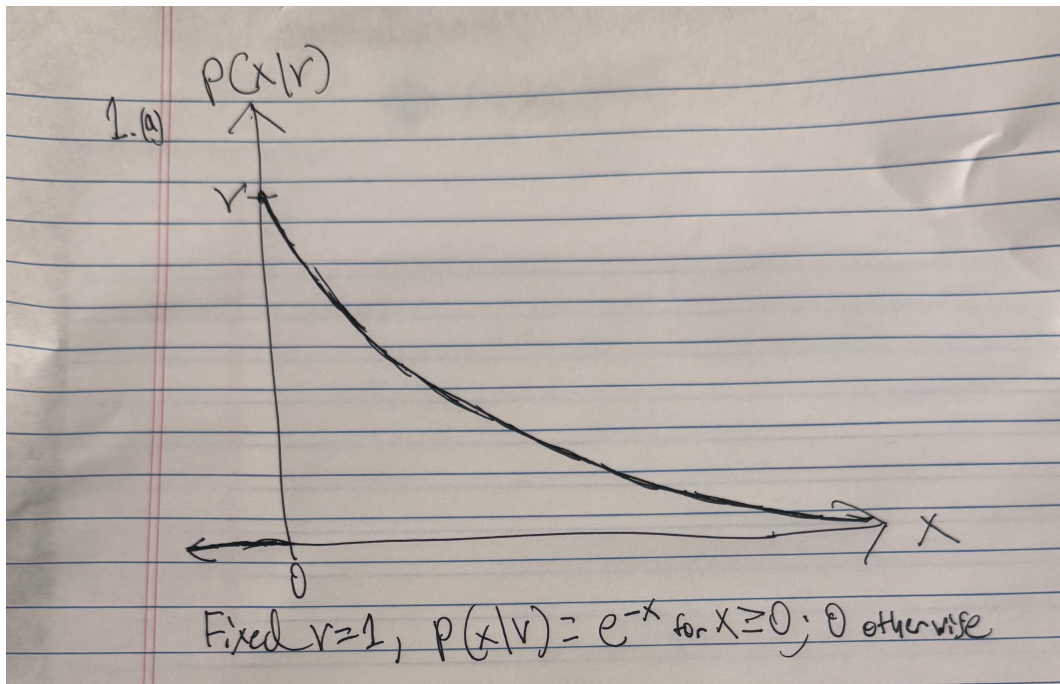
1.

Let X have an exponential distribution

$$p(x|v) = \begin{cases} ve^{-vx}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

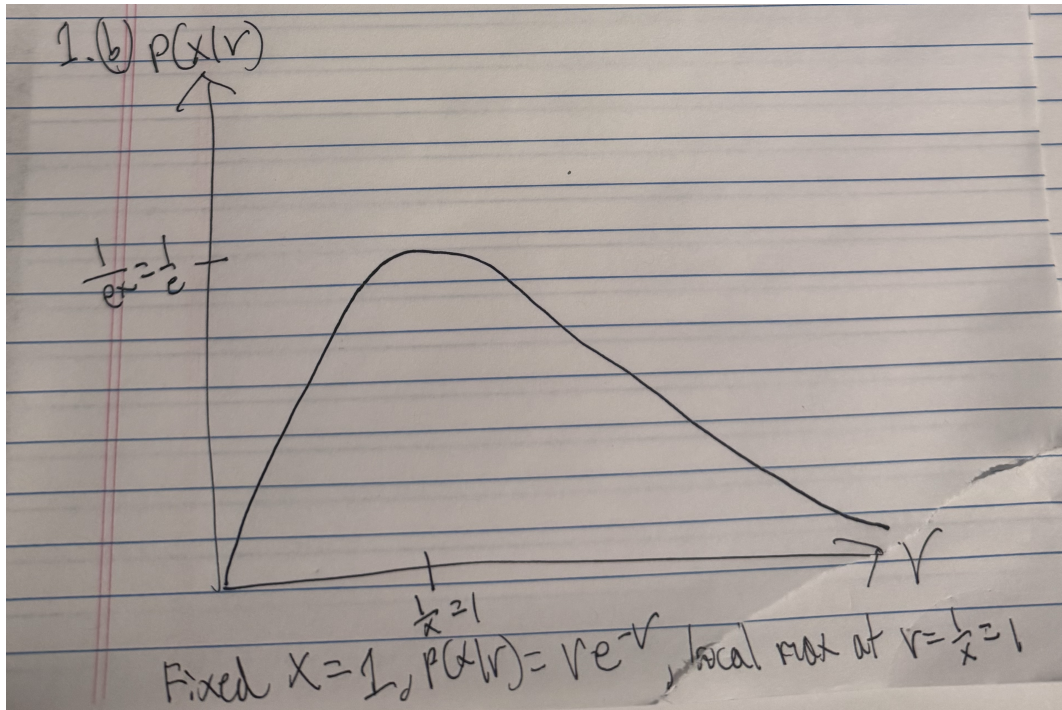
(a)

Sketch $p(x|v)$ versus x for a fixed value of the parameter v .



(b)

Sketch $p(x|v)$ versus v , $v > 0$ for a fixed value of x .



(c)

Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|v)$. Show that the maximum likelihood estimate for v is given by

$$\hat{v} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}$$

We can derive the MLE for v directly. We first write the likelihood function. Suppose $x_1, \dots, x_n$ are independent observations drawn from the exponential distribution, as defined by $p(x|v)$; we just multiply for a likelihood function of

$$L(v) = \prod_{k=1}^n v e^{-v x_k}$$

We then take the log likelihood:

$$\begin{aligned} l(v) &= \ln\left(\prod_{k=1}^n v e^{-v x_k}\right) \\ &= \sum_{k=1}^n \ln(v e^{-v x_k}) \\ &= \sum_{k=1}^n (\ln(v) + \ln(e^{-v x_k})) \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n (\ln(v) - vx_k) \\
&= n \ln(v) - \sum_{k=1}^n vx_k
\end{aligned}$$

and differentiate:

$$\begin{aligned}
\frac{dl}{dv} &= \frac{d}{dv} (n \ln(v) - \sum_{k=1}^n vx_k) \\
&= \frac{n}{v} - \sum_{k=1}^n x_k
\end{aligned}$$

Verifying this is a maximum by taking the second derivative:

$$\begin{aligned}
\frac{d^2l}{dv^2} &= \frac{d}{dv} \left(\frac{n}{v} - \sum_{k=1}^n x_k \right) \\
&= -\frac{n}{v^2} - 0 = -\frac{n}{v^2}
\end{aligned}$$

which we see is indeed negative as both the numerator $n > 0$ and the denominator is also always positive; multiplying by this then gives us a negative value, so this is indeed a maximum. Finally, set equal to zero to solve:

$$\begin{aligned}
0 &= \frac{n}{v} - \sum_{k=1}^n x_k \\
\sum_{k=1}^n x_k &= \frac{n}{v} \\
v &= \frac{n}{\sum_{k=1}^n x_k}
\end{aligned}$$

So we get

$$\hat{v} = \frac{n}{\sum_{k=1}^n x_k} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}$$

exactly as we sought to find.

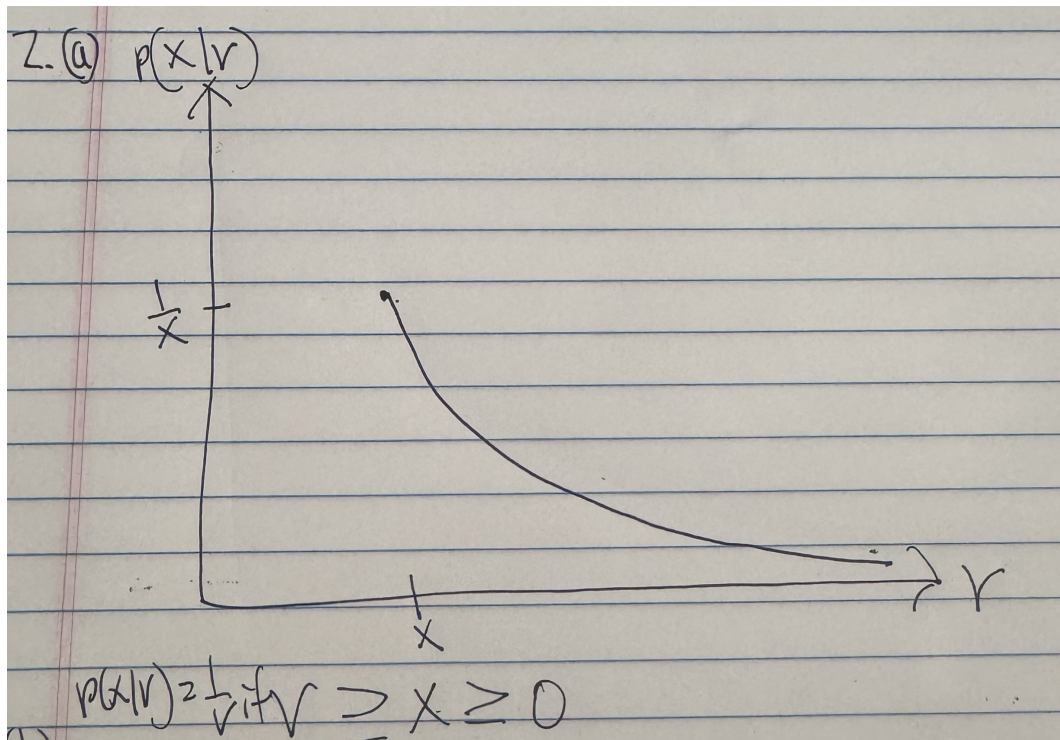
2.

Let X have a uniform distribution

$$p(x|v) = \begin{cases} \frac{1}{v}, & 0 \leq x \leq v \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Sketch $p(x|v)$ versus v for an arbitrary value of x .



(b)

Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|v)$. Show that the maximum likelihood estimate for v is $\max_k x_k$.

By definition, we have that

$$L(v) = \prod_{k=1}^n p(x_k|v)$$
$$L(v) = \begin{cases} \frac{1}{v^n}, & v \geq \max_k(x_k) \\ 0, & \text{otherwise.} \end{cases}$$

We note that $\frac{1}{v^n}$ continuously decreases as v increases; we therefore must choose the smallest possible value of v . We also have the restriction that v must be at least as large as the maximum x_k value, or $\max_k x_k$. We therefore have that

$$\hat{v} = \max_k x_k$$

exactly as we sought to show.

(c)

Find the method of moments estimator for v .

Let $X_1, \dots, X_n \sim \text{Uniform}(0, v)$. We therefore have the first population moment as

$$\begin{aligned}\alpha_1 = E[X] &= \int_0^v x \cdot \frac{1}{v} dx \\ &= \frac{1}{v} \cdot \frac{x^2}{2} \Big|_0^v \\ &= \frac{1}{v} \cdot \frac{v^2}{2} - 0 \\ E[X] &= \frac{v}{2}\end{aligned}$$

We would then also have the corresponding first sample moment as

$$\hat{\alpha}_1 = \frac{1}{n} \sum_{k=1}^n x_k = \bar{x}$$

So we finally take the method of moments estimator to be

$$\frac{\hat{v}}{2} = \bar{x}$$

$$\boxed{\hat{v} = 2\bar{x}}$$

3.

4.

(a)

(b)

5.

6.

(a)

(b)

(c)